

SECSR2043 OPERATING SYSTEMS

[20 Marks]

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Marks

Instruction: Please answer all of the following questions. Whenever the 🖐 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *.; cp ../fruits/*. *.;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]

\$HOME

```
GNU nano 7.2 farra.sh *
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*. *.; cp ../fruits/*. *.; cp ../*.jar .|
```

```

farra@secr2043:~/lab3$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   '-- proton.c
|-- dates
|   |-- appointment
|   '-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   '-- proton.c
|-- farra.sh
|-- fruits
|   |-- apple.txt
|   '-- orange.txt
'-- helloworld.jar

```

- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

```

#!/bin/bash
echo "Enter file extension:"
read ext
echo "Files with .$ext extension:"
ls *.$ext
count=$(ls *.$ext | wc -l)
echo "Number of .$ext files: $count"

```

```

farra@secr2043:~$ bash filecounter.sh
Enter file extension:
c
Files with .c extension:
forkWait.c  mainprocess.c  subprocess1.c  subprocess2.c
Number of .c files: 4

```

2. The following Figure 1 illustrates a tree structure of some directories and files.

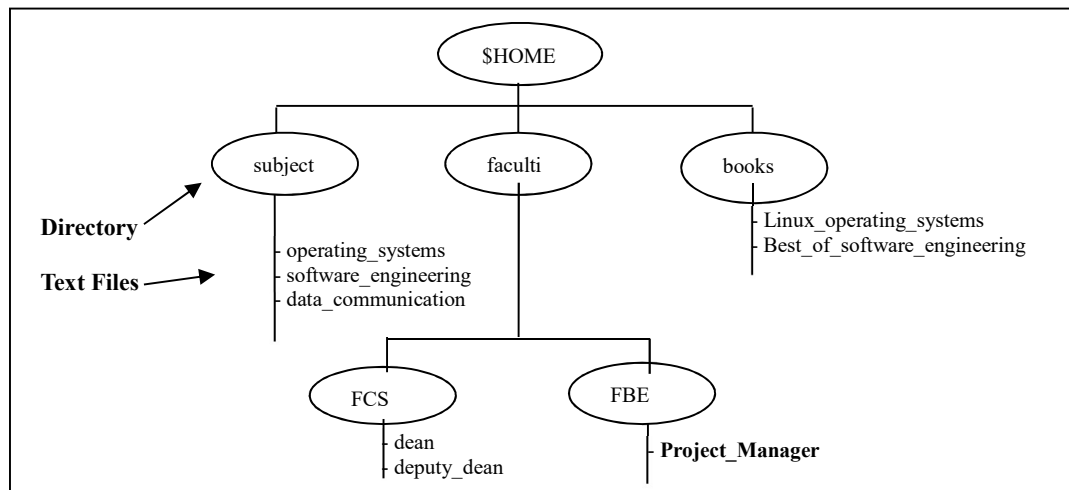


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

```

GNU nano 7.2                               myname2a.sh *
#!/bin/bash

mkdir -p $HOME/subject
mkdir -p $HOME/faculti/FCS
mkdir -p $HOME/faculti/FBE
mkdir -p $HOME/books

cd $HOME/subject
touch operating_systems software_engineering data_communication

cd $HOME/books
echo "Linux operating systems" > Linux_operating_systems
echo "Best of software engineering" > Best_of_software_engineering

cd $HOME/faculti/FBE
echo "Project Manager" > Project_Manager

cd $HOME/faculti/FCS
echo "dean" > dean
echo "deputy dean" > deputy_dean

```

```

farra@secr2043:~/lab3$ ./myname2a.sh
farra@secr2043:~/lab3$ ls -R
.:
books  dates  faculti  filecounter.sh  helloworld.jar  subject
cars   drinks  farra.sh  fruits          myname2a.sh

./books:
Best_of_software_engineering  Linux_operating_systems

./cars:
accord.c  civic.c  proton.c

./dates:
appointment  dateoftheday

./drinks:
accord.c  apple.txt  civic.c  helloworld.jar  orange.txt  proton.c

./faculti:
FBE  FCS

./faculti/FBE:
Project_Manager

./faculti/FCS:
dean  deputy_dean

./fruits:
apple.txt  orange.txt

./subject:
data_communication  operating_systems  software_engineering

```

- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.
- [2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--

- c) Write another bash script (called myname2c.sh) that will change the access

control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓



```
GNU nano 7.2 myname2c.sh *
#!/bin/bash

chmod 755 subject
chmod 511 books/Best_of_software_engineering
chmod 775 faculti/FCS
chmod 644 faculti/FBE/Project_Manager
```

```
farra@secr2043:~/lab3$ chmod +x myname2c.sh
farra@secr2043:~/lab3$ ./myname2c.sh
farra@secr2043:~/lab3$ ls -lR
```

```
-rwxr-xr-x 1 farra farra 136 Jun 19 13:04 myname2c.sh
drwxr-xr-x 2 farra farra 4096 Jun 19 07:44 subject
```

```
./books:
total 8
-r-x--x--x 1 farra farra 29 Jun 19 07:44 Best_of_software_engineering
-rw-r--r-- 1 farra farra 24 Jun 19 07:44 Linux_operating_systems
```

```
./faculti:
total 8
drwxr-xr-x 2 farra farra 4096 Jun 19 07:44 FBE
drwxrwxr-x 2 farra farra 4096 Jun 19 07:44 FCS
```

```
./faculti/FBE:
total 4
-rw-r--r-- 1 farra farra 16 Jun 19 07:44 Project_Manager
```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr-xr-x
Best_of_software_engineering	-r-x--x--x
FCS	drwxrwxr-x
project_manager	--rw-r--r--

End of Lab 3

**** All the Best for Final Exam ****